

Pranjal Shukla

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Education

The University Of Edinburgh, Scotland, UK

September 2022 - May 2027

Integrated M.Eng. (Hons.) Electronics with Computer Science **[1st Class : 76%]**

Relevant Courses - Functional Programming, Object-Oriented Programming, Computer Architecture and Systems, Algorithms and Data Structures, Informatics Large Practical, Operating Systems, Compiling Techniques, Computer Security, Machine Learning Practical, Software Testing, and Software and Embedded Systems.

Work Experience

Kaiteki AI | Remote Role, Saratoga, California

June 2025 - September 2025

SWE Intern

- Built a background Android service for automated screenshot capture and OCR. Over time reduced runtime overhead by 35% through optimized image handling using compressed ImageBitmap, which only reduced the accuracy by 7-11%. Instead of performing the OCR on a (.png), capturing the bitmap directly helped in reducing the overhead of conversion.
- Integrated AI-based profanity/mental health classifier, triggering secure alerts with a latency of about ~400ms. Detection of activities such as bullying, harassment, profane language, vulgarity, etc triggered an alert to the group host.
- Optimized image handling, text extraction, and background execution for minimal performance impact. Instead of continuous running of the OCR and classification service, batch servicing helped in reducing the computation costs.
- Creating a service that monitors incoming messages and texts from different social media apps for achieving the similar results in sandboxed environments such as iOS.

Tech Mahindra Makers Lab | Pune, India

July 2023 - September 2023

SWE Intern

- Designed a responsive UI (React and Tailwind) which in turn boosted forum engagement by 15%.
- Collaborated on the development of an intelligent, real-time chat assistant to answer user queries. Built a real-time chat assistant using Django and React) leveraging APIs/state management that delivered instant query responses.
- Explored foundational deep learning concepts using TensorFlow and Keras, experimenting with text classification models and gaining insights into model training, evaluation, and tuning.
- Explored NLP basics by experimenting with different libraries - learned about text-to-speech, language/grammar, and transformer modelling.

Projects

FLOW-UKT(Uncertainty Knowledge Tracing)

April 2026

- Developed a novel flow-based approach for knowledge tracing and demonstrated performance comparable to state-of-the-art baselines, while improving the modelling of predictive uncertainty.
- Built an uncertainty-aware knowledge tracing system in Python and PyTorch, implementing FlowUKT with normalizing flows alongside UKT and MC-Dropout baselines for next-response prediction of the student, Included training, evaluation, and plotting pipelines on ASSIST2009 benchmark dataset(123 skills), including config-driven experiments, ablation sweeps, and result summaries.
- Achieved up to 0.829 validation AUC and 0.817 test AUC with annealed FlowUKT, the matched no-annealing baseline levelled at 0.630 validation AUC. The state-of-the-art model has a test AUC of 0.851.

Quantitative Sports Betting [G-Research Runner Up] | HackTheBurgh

November 2025

- Created a real-time NBA betting terminal that leveraged live game stats and 77 bookmaker odds for over 6000 unique categories of bets, live headlines, modelled expected value insights, and risk management for 4800 unique basketball prop bets.
- Created multiple ML and statistical models using 18 input features to predict multiple distributions, such as position, negative binomial, etc., of player stats. The model achieved an ROC-AUC score of 0.87. The training set was of 880,000 games.

- Multi-threaded the ML and the statistical models and shared cache weights to cut down model prediction latency from 12 minutes to 30 seconds.

Optiver Trading Challenge [4th position] | HackTheBurgh

November 2025

- Created a high-performance multi-threaded trading system combining dual-listing arbitrage, market-making, and news-driven momentum strategies using Python and the Optibook API.
- Exploited concurrent execution, thread-safe rate-limiting, and adaptive quote management to achieve sub-second latency in trades.
- Built a custom sentiment analysis engine using lexical expansion from training data to optimize PnL stability and execution speed. The sentiment analysis was performed on dummy news that was being generated every few seconds on the Optibook terminal. Using it we tweaked the trade decision by incorporating the sentiment score.

Code For Good [AI search tool]

October 2025

- The aim of the project was to make it easy for the NPO to reach out, maintain accurate information with different organizations so that users of the NPO services can easily access and inquire about the services they need. So in essence the main issues were - maintainability, access, accuracy.
- To solve this I created a lightweight RAG model with vectorised search on the database to find appropriate organization names for user services. The backend and database was handled by PocketBase with API function enabled.
- For maintainability created an easy to use admin database page where the organizations could be verified and events could be edited.
- The front-end website used ReactJS, TailwindCSS and React Native to make it compatible for mobile devices as well. Smart extraction using LLM integration for minimal work by organizations.
- Added functionality was to send automated reminder emails to organizations to keep their events and services up to date. A TTL(attached to each organization object) to overcome the issue of misinformation to the users.

SpringBoot API Service [Pizza Drone Controller]

January 2025

- Created an API service using SpringBoot to control a pizza drone service, it processed orders, fetched no-fly zone areas and used A-star path finding algorithms to find the quickest path.
- Extensive testing using JUnit tests of all the different services created. Used Postman and curl as well to test/verify various different endpoints of the API service.
- A-Star path finding algorithm was one of the quickest with the average response time(~400 ms). It could also provide a geo-spatial route in JSON format that could be visualized easily using any map markers. It used pre-computed heuristics to improve the computation of the path.

Technical Skills

Proficient - Java(SpringBoot), Python (NumPy, PyTorch, Matplotlib, Pandas, Flask), HTML, C++, LTSpice, CSS, MS Office, Tailwind CSS, LaTeX, LINUX, SQL, Swift, Android Studios

Intermediate - CAD 3D modeling, MatLab, STM32 CubeIDE, Vivado, Xilinx(Verilog), Haskell, MIPS Assembly, JavaScript(ReactJS, NodeJS, ExpressJS)